

Marcus Ortyl

AI Engineer | Full-Stack Engineer

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Profile

Full-stack engineer with 3+ years' experience shipping production web applications, now specialising in LLM-powered products. Hands-on with retrieval-augmented generation (RAG), prompt engineering, knowledge-grounded AI systems and multi-model orchestration on the Anthropic Claude API. Strong product mindset, building end to end across Python/FastAPI backends and React/Next.js frontends.

Core Skills

AI / LLM: Anthropic Claude API, RAG & knowledge grounding, embeddings & vector search, evals & AI guardrails, prompt engineering, multi-model orchestration, structured output generation

Backend: Python, FastAPI, Node.js, REST APIs, PostgreSQL, SQLite, Redis, Prisma

Frontend: TypeScript, JavaScript, React, Next.js, Tailwind CSS, WebGL / PIXI.js

Tools & Other: Git, GitHub, Vercel, AWS, OAuth, Agile

AI Engineering

Capstan – github.com/Mortyl/capstan – getcapstan.vercel.app

- Built and deployed a live multi-tenant SaaS that uses AI to draft customer-support replies, grounded in each customer's own help docs and cited back to the exact source
- Engineered the full retrieval pipeline: chunking and embedding docs (Voyage AI) into PostgreSQL/pgvector, with Claude generating answers and durable background jobs (Inngest) handling ingestion and indexing at scale
- Added retrieval-confidence guardrails and an automated eval suite in CI to keep AI answers accurate, escalating weak matches to a human instead of guessing
- Designed the full commercial layer, not just the AI: org-scoped multi-tenancy with row-level data isolation, Auth.js authentication and Stripe subscription billing
- Shipped to production solo (Next.js/TypeScript, Prisma, Neon Postgres, Vercel) and verified with end-to-end smoke tests against the live deployment, including the ingestion pipeline

Professional Experience

10/2024 - Present

Freelance Web Developer – *Remote*

- Delivered full-stack web applications for multiple clients using React, Next.js, TypeScript and Node.js, improving scalability and maintainability
- Rebuilt legacy applications with modern frameworks, reducing page load times by 60% and significantly improving UX and SEO performance
- Redesigned and optimised a payment flow, reducing checkout friction and increasing successful transaction completion rates

04/2023 - 04/2024

Application Developer, Illumo Digital – Bristol

- Built and delivered production-ready features for a React-based sports-tech platform used in professional rugby and UFC environments
- Improved application performance and stability through targeted bug fixes and optimisations across React and Node.js codebases
- Enhanced mobile usability and responsiveness via React Native improvements

03/2021 - 11/2022

Software Engineer, Brightpearl – Bristol

- Developed AWS-based e-commerce platform features within cross-functional agile teams
- Contributed to migration from PHP monolith to Java microservices using Kubernetes
- Reduced production bugs by 20% by improving unit test coverage and enhancing API testing workflows

11/2019 - 01/2021

Application Developer, ALD Automotive – Bristol

- Built and maintained a finance web application using Angular and C# in an agile development team
- Developed reusable frontend components, improving UI consistency and efficiency
- Wrote production SQL queries against live financial databases supporting daily operations

Projects

PoEProfessor – github.com/Mortyl/poe-professor

- Designed and implemented a knowledge-grounded AI assistant (RAG) using the Anthropic Claude API and a curated knowledge base of 80+ documents with keyword-triggered context injection, reducing hallucinations in a rapidly changing game ecosystem
- Orchestrated multiple Claude models for cost and quality: Sonnet for the reasoning-heavy chat companion and Haiku for fast, structured build generation
- Processed and analysed 20,000+ player builds through automated data collection pipelines, generating statistical recommendations for AI-generated build guides

AI Dungeon-master – github.com/Mortyl/ai-dungeon-master

- Agentic tool-using AI (Claude API): the LLM proposes structured tool calls, a deterministic Python engine validates every action, with a CI eval harness asserting agent correctness

Education

2018 - 2019

DevelopMe, Bristol – Web Developer Bootcamp

2014 - 2018

University of the West of England, Bristol – BSc Biological Science (2:1)